Causal normalizing flows: from theory to practice

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Abstract

In this work, we deepen on the use of normalizing flows for causal inference. Specifically, we first leverage recent results on non-linear ICA to show that causal models are identifiable from observational data given a causal ordering, and thus can be recovered using autoregressive normalizing flows (NFs). Second, we analyse different design and learning choices for *causal normalizing flows* to capture the underlying causal data-generating process. Third, we describe how to implement the *do-operator* in causal NFs, and thus, how to answer interventional and counterfactual questions. Finally, in our experiments, we validate our design and training choices through a comprehensive ablation study; compare causal NFs to other approaches for approximating causal models; and empirically demonstrate that causal NFs can be used to address real-world problems—where mixed discrete-continuous data and partial knowledge on the causal graph is the norm. The code for this work can be found at https://github.com/psanch21/causal-flows.

1 Introduction

Deep learning is increasingly used for causal reasoning, that is, for finding the underlying causal relationships among the observed variables (*causal discovery*), and answering *what-if* questions (*causal inference*) from available data [30]. Our focus in this paper is to solve causal inference problems using only observational data and (potentially partial) knowledge on the causal graph of the underlying structural causal model (SCM). This is exemplified in Fig. 1, where our proposed framework is able to estimate the (unobserved) causal effect of externally intervening on the sensitive attribute (red and yellow distributions), *using solely observed data (blue distribution) and partial information about the causal relationship between features*.

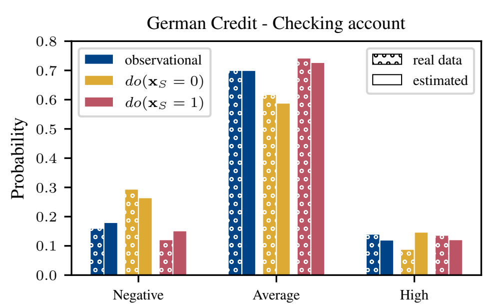

Figure 1: Observational and interventional distributions of the categorical variable *checking account* of the German Credit dataset [9], and their estimated values according to a causal normalizing flow. $\mathbf{x}_S$ is a binary variable representing the users' sex.

In this context, previous works have mostly relied on different deep neural networks (DNNs)—e.g., normalizing flows (NFs) [1, 25, 28, 29], generative adversarial networks (GANs) [22, 41], variational autoencoders (VAEs) [17, 42], Gaussian processes (GPs) [17], or denoising diffusion probabilistic models (DDPMs) [3]—to iteratively estimate the conditional distribution of each observed variable given its causal parents, thus using an independent DNN per observed variable. Hence, to predict the effect of an intervention in the causal data-generating process, these approaches fix the value of the intervened variables when computing the new value for their children. However, they may also suffer

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from error propagation—which worsens with long causal paths—and a high number of parameters, which is addressed in practice with ad-hoc parameter amortization techniques [29]. Moreover, several approaches also rely on implicit distributions [3, 22, 29, 34], and thus do not allow evaluating the learnt distribution.

In contrast, and similar to [18, 34, 36, 43], we here aim at learning the full causal-generating process using a single DNN and, in particular, using a *causal normalizing flow*. To this end, we first theoretically demonstrate that causal NFs are a natural choice to approximate a broad class of causal data-generating processes (§3). Then, we design causal NFs that inherently satisfy the necessary conditions to capture the underlying causal dependencies (§4), and introduce an implementation of the do-operator that allows us to efficiently solve causal inference tasks (§5). Importantly, our causal NF framework allows us to deal with mixed continuous-discrete data and partial knowledge on the causal graph, which is key for real-world applications. Finally, we empirically validate our findings and show that causal NFs outperform competing methods also using a single DNN to approximate the causal data-generating process (§6).

Related work To the best of our knowledge, the closest works to ours are [18, 36, 43], as they all capture the whole causal data-generating process using a single DNN. Our approach generalizes the result from Khemakhem et al. [18], which also relies on autoregressive normalizing flows (ANF), but only considers affine ANFs and data with additive noise. In contrast, our work provides a tighter connection between ANFs and SCMs (affine or not), more general identifiability results, and sound ways to both embed causal knowledge in the ANF, and to apply the do-operator. Another relevant line of works connect SCMs with GNNs [34, 43], and despite making little assumptions on the underlying SCM, they lack identifiability guarantees, and interventions on the GNN are performed by severing the graph, which we show in App. C may not work in general. Nevertheless, it is worth noting that the way we use $\boldsymbol{A}$ in the network design (§4) is inspired by these works. Finally, contemporary works have also proposed to use $\boldsymbol{A}$ to mask the connections of an ANF [1, 12, 28], albeit without the general theoretical results and design characterization provided in this work.

2 Preliminaries and background

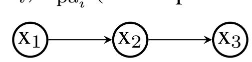
2.1 Structural causal models, interventions, and counterfactuals

A structural causal model (SCM) [30] is a tuple $\mathcal{M} = (\tilde{\mathbf{f}}, P_{\mathbf{u}})$ describing a data-generating process that transforms a set of d exogenous random variables, $\mathbf{u} \sim P_{\mathbf{u}}$, into a set of d (observed) endogenous random variables, $\mathbf{x}$, according to $\tilde{\mathbf{f}}$. Specifically, the endogenous variables are computed as follows:

$$\mathbf{u} \coloneqq (\mathrm{u}_1, \mathrm{u}_2, \dots, \mathrm{u}_d) \sim P_{\mathbf{u}}\,, \qquad \mathrm{x}_i = \tilde{f}_i(\mathbf{x}_{\mathrm{pa}_i}, \mathrm{u}_i)\,, \qquad \text{for } i = 1, 2, \dots, d\,. \tag{1}$$

In other words, each i-th component of $\tilde{\mathbf{f}}$ maps the i-th exogenous variable u_i to the i-th endogenous variable x_i, given the subset of the endogenous variables that directly cause x_i, $\mathbf{x}_{\mathrm{pa}_i}$ (causal parents).

An SCM also induces a causal graph, a powerful tool to reason about the causal dependencies of the system. Namely, the causal graph of an SCM $\mathcal{M} = (\tilde{\mathbf{f}}, P_{\mathbf{u}})$ is the directed graph that describes the functional dependencies of the causal mechanism. We can define the *adjacency matrix of the causal graph* as $\boldsymbol{A} \coloneqq \nabla_{\mathbf{x}}\tilde{\mathbf{f}}(\mathbf{x}, \mathbf{u}) \neq \mathbf{0}$, where $\mathbf{0}$ is the constant zero function, and the comparisons are made elementwise. Furthermore, the direct causes of the i-th variable (pa_i in Eq. 1) are the *parent* nodes of the i-th node in $\boldsymbol{A}$, and the *ancestors* of this node (which we denoted by an_i) are its (in)direct causes. See Fig. 2 for an example of a causal chain.



$$\pi = (1 \quad 2 \quad 3)$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Figure 2: Causal graph, and its causal ordering π and adjacency matrix $\boldsymbol{A}$.

In the case that $\boldsymbol{A}$ is acyclic, we can pick a causal ordering describing which variables do *not* cause others, and which ones *may* cause them. Namely, a permutation π is said to be a *causal ordering* of an SCM $\mathcal{M}$ if, for every x_i that directly causes x_j, we have $\pi(i) < \pi(j)$. Note that this definition equals that of a topological ordering and, without loss of generality, we will assume throughout this work that the variables are sorted according to a causal ordering.

Importantly, besides describing the (observational) data-generating process, SCMs enable *causal inference* by allowing us to answer *what-if* questions regarding: i) how the distribution over the observed variables would be if we force a fixed value on one of them (*interventional queries*); and ii) what would have happened to a specific observation, if one of its dimensions would have taken a different value (*counterfactual queries*).

Structural equivalence To reason about causal dependencies, we introduce the notion of structural equivalence. We say that two matrices $\boldsymbol{S}$ and $\boldsymbol{R}$ are *structurally equivalent*, denoted $\boldsymbol{S} \equiv \boldsymbol{R}$, if both matrices have zeroes exactly in the same positions. Similarly, we say that $\boldsymbol{S}$ is *structurally sparser* than $\boldsymbol{R}$, denoted as $\boldsymbol{S} \preceq \boldsymbol{R}$, if whenever an element of $\boldsymbol{R}$ is zero, the same element of $\boldsymbol{S}$ is zero.

2.2 Autoregressive normalizing flows

Normalizing flows (NFs) [27] are a model family that express the probability density of a set of observations using the change-of-variables rule. Given an observed random vector $\mathbf{x}$ of size d, a normalizing flow is a neural network with parameters $\boldsymbol{\theta}$ that takes $\mathbf{x}$ as input, and outputs

$$T_{\boldsymbol{\theta}}(\mathbf{x}) =: \mathbf{u} \sim P_{\mathbf{u}} \quad \text{with log-density} \quad \log p(\mathbf{x}) = \log p(T_{\boldsymbol{\theta}}(\mathbf{x})) + \log|\det(\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x}))|\,, \tag{2}$$

where $P_{\mathbf{u}}$ is a base distribution that is easy to evaluate and sample from. Since Eq. 2 provides the log-likelihood expression, it naturally leads to the use of maximum likelihood estimation (MLE) [2] for learning the network parameters $\boldsymbol{\theta}$. While many approaches have been proposed in the literature [27], here we focus on autoregressive normalizing flows (ANFs) [20, 26]. Specifically, in ANFs the i-th output of each layer l of the network, denoted by z_i^l, is computed as

$$\mathrm{z}_i^l := \tau_i^l(\mathrm{z}_i^{l-1}; \mathbf{h}_i^l)\,, \quad \text{where} \quad \mathbf{h}_i^l := c_i^l(\mathbf{z}_{1:i-1}^{l-1})\,, \tag{3}$$

and where τ_i and c_i are termed the transformer and the conditioner, respectively. The transformer is a strictly monotonic function of z_i^{l-1}, while the conditioner can be arbitrarily complex, yet it only takes the variables preceding z_i as input. As a result, ANFs have triangular Jacobian matrices, $\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x})$.

3 Causal normalizing flows

Problem statement Assume that we have a sequence of i.i.d. observations $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\}$ generated according to an unknown SCM $\mathcal{M}$, from which we have partial knowledge of its causal structure. Specifically, we know at least its causal ordering π, and at most the whole causal graph $\boldsymbol{A}$. Our objective in this work is to design and learn an ANF $T_{\boldsymbol{\theta}}$, with parameters $\boldsymbol{\theta}$, that captures $\mathcal{M}$ by maximizing the observational likelihood (MLE), i.e.,

$$\underset{\boldsymbol{\theta}}{\text{maximize}} \; \frac{1}{N} \sum_{n=1}^{N} \Big[\log p\left(T_{\boldsymbol{\theta}}\left(\boldsymbol{x}_n\right)\right) + \log|\det(\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}\left(\boldsymbol{x}_n\right))| \Big]\,, \tag{4}$$

and that can successfully answer interventional and counterfactual queries during deployment, thus enabling causal inference. We refer to such a model as a *causal normalizing flow*.

Assumptions We restrict the class of SCMs considered by making the following fairly common assumptions: i) *diffeomorphic data-generating process*, i.e., $\tilde{\mathbf{f}}$ is invertible, and both $\tilde{\mathbf{f}}$ and its inverse are differentiable; ii) *no feedback loops*, i.e., the induced causal graph is acyclic; and iii) *causal sufficiency*, i.e., the exogenous variables are mutually independent, $p(\mathbf{u}) = \prod_i p(\mathbf{u}_i)$.

SCMs as TMI maps To achieve our objective, and bridge the gap between SCMs and normalizing flows, we resort to triangular monotonic increasing (TMI) maps, which are autoregressive functions whose i-th component is strictly monotonic increasing with respect to its i-th input. TMI maps hold a number of useful properties, such as being closed under composition and inversions. Conveniently, a layer of an ANF (Eq. 3) is a parametric TMI map that can approximate any other TMI map arbitrarily well, which makes ANFs also TMI maps approximators;[2] a fact that has been exploited in the past to prove that ANFs are universal density approximators [27].

We now show that any SCM can be rewritten as a tuple $(\mathbf{f}, P_{\mathbf{u}}) \in \mathcal{F} \times \mathcal{P}_{\mathbf{u}}$, where $\mathcal{F}$ is the set of all TMI maps, and $P_{\mathbf{u}}$ is the set of all fully-factorized distributions, $p(\mathbf{u}) = \prod_i p(\mathbf{u}_i)$. First, given an acyclic SCM $\mathcal{M} = (\tilde{\mathbf{f}}, P_{\mathbf{u}})$ with $\tilde{\mathbf{f}} : \mathbb{X} \times \mathbb{U} \to \mathbb{X}$ as in Eq. 1, we can always unroll $\tilde{\mathbf{f}}$ by recursively replacing each x_i in the causal equation by its function $\tilde{f}_i$ (see Fig. 4b for an example), obtaining an equivalent non-recursive function $\hat{\mathbf{f}} : \mathbb{U} \to \mathbb{X}$. This function $\hat{\mathbf{f}}$ writes each x_i as a function of its exogenous ancestors $\mathbf{u}_{\mathrm{an}_i}$ and, since $\mathcal{M}$ is acyclic, $\hat{\mathbf{f}}$ is a triangular map. For simplicity, assume that $P_{\mathbf{u}}$ is a standard uniform distribution. Then, following the causal ordering, we can apply a Darmois

[2] While it is a common to shuffle the inputs for each layer, we keep the same order across the network.

construction [8, 15] and replace each function $\hat{f}_i$ by the conditional quantile function of the variable x_i given $\mathbf{x}_{\mathrm{pa}_i}$ (which depends on $\mathbf{u}_{\mathrm{an}_i}$) eventually arriving to a TMI map $\mathbf{f}$. This procedure follows the proof for non-identifiability in ICA [15], but restricted to one ordering. The case for a general $P_{\mathbf{u}}$ follows a similar construction, but using a Knöthe-Rosenblatt (KR) transport [21, 33] instead.

Isolating the exogenous variables Now that we have SCMs and causal NFs under the same family class—i.e., the family $\mathcal{F} \times \mathcal{P}_{\mathbf{u}}$ of TMI maps with fully-factorized distributions—we leverage existing results on identifiability to show that we can find a causal NF $T_{\boldsymbol{\theta}}$ such that the i-th component of $T_{\boldsymbol{\theta}}(\mathbf{x})$ is a function of the true exogenous variable u_i that generated the observed data. More precisely, note that, since we can rewrite $\mathcal{M}$ as an element of the family $\mathcal{F} \times \mathcal{P}_{\mathbf{u}}$, identifying the true exogenous variables of an SCM $\mathcal{M}$ is equivalent to solving a non-linear ICA problem with TMI generators, for which Xi and Bloem-Reddy [40] proved the following (re-stated to match our setting):

Theorem 1 (Identifiability)**.** If two elements of the family $\mathcal{F} \times \mathcal{P}_{\mathbf{u}}$ (as defined above) produce the same observational distribution, then the two data-generating processes differ by an invertible, component-wise transformation of the variables $\mathbf{u}$.

Thm. 1 implies that, if we can find a causal NF $(T_{\boldsymbol{\theta}}, P_{\boldsymbol{\theta}}) \in \mathcal{F} \times \mathcal{P}_{\mathbf{u}}$ that matches the observational distribution generated by $\mathcal{M} = (\mathbf{f}, P_{\mathcal{M}}) \in \mathcal{F} \times \mathcal{P}_{\mathbf{u}}$, then we know that the exogenous variables of the flow differ from the real ones by a function of each component independently, i.e., $T_{\boldsymbol{\theta}}(\mathbf{f}(\mathbf{u})) \sim P_{\boldsymbol{\theta}}$ with $\mathbf{u} \sim P_{\mathcal{M}}$ and $T_{\boldsymbol{\theta}}(\mathbf{f}(\mathbf{u})) = \boldsymbol{h}(\mathbf{u}) = (h_1(\mathrm{u}_1), h_2(\mathrm{u}_2), \ldots, h_d(\mathrm{u}_d))$, where each h_i is an invertible function. Fig. 3 graphically illustrates Thm. 1. Furthermore, Thm. 1 also implies that the functional dependencies of the causal NF must agree with that of the SCM, i.e., that $T_{\boldsymbol{\theta}}$ needs to be *causally consistent* with $\mathcal{M}$. We formally present this result in the following corollary (proof can be found in App. A), where $\boldsymbol{I}$ denotes the identity matrix:

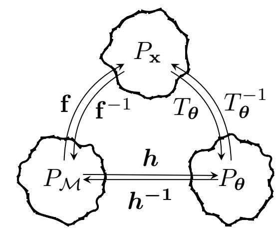


Figure 3: Thm. 1 as a commutative diagram

Corollary 2 (Causal consistency)**.** If a causal NF $T_{\boldsymbol{\theta}}$ isolates the exogenous variables of an SCM $\mathcal{M}$, then $\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x}) \equiv \boldsymbol{I} - \boldsymbol{A}$ and $\nabla_{\mathbf{u}} T_{\boldsymbol{\theta}}^{-1}(\mathbf{u}) \equiv \boldsymbol{I} + \sum_{n=1}^{\mathrm{diam}(\boldsymbol{A})} \boldsymbol{A}^n$, where $\boldsymbol{A}$ is the causal adjacency matrix of $\mathcal{M}$. In other words, $T_{\boldsymbol{\theta}}$ is causally consistent with the true data-generating process, $\mathcal{M}$.

A sketch of the proof goes as follows: since Fig. 3 is a commutative diagram, we can write the result of $T_{\boldsymbol{\theta}}$ and $T_{\boldsymbol{\theta}}^{-1}$ in terms of the true $\mathbf{f}$, $\boldsymbol{h}$, and their inverses. Then, we can use the chain rule to compute their Jacobian matrices, and since $\boldsymbol{h}$ has a diagonal Jacobian matrix, it preserves the structure of the Jacobian matrices of $\mathbf{f}$ and its inverse. To sum up, we have shown that *causal NFs are a natural choice to estimate an unknown SCM* by showing that: i) both SCMs and causal NFs fall within the same family $\mathcal{F} \times \mathcal{P}_{\mathbf{u}}$; ii) any two elements of this family with identical observational distributions are causally consistent; and iii) they differ by an invertible component-wise transformation.

3.1 Causal NFs for real-world problems

To bring theory closer to practice, we need to extend causal NFs to handle mixed discrete-continuous data and partial knowledge on the causal graph, which are common properties of real-world problems. Due to space limitations, we provide here a brief explanation, and formalize these ideas in App. A.2.

Discrete data To extend our results to also account for discrete data, we take advantage of the general model considered by Xi and Bloem-Reddy [40] that includes observational noise (independent of the exogenous variables), and consider a continuous version of the observed discrete variables by adding to them independent noise $\varepsilon \in [0, 1]$ (e.g., from a standard uniform), such that the real distribution is still recoverable. Intuitively, our approach assumes that discrete variables correspond to the integer part of (noisy) continuous variables generated according to an SCM fulfilling our assumptions, such that both our theoretical and practical insights still apply.

Partial knowledge While we rarely know the entire causal graph $\boldsymbol{A}$, we often have a good grasp on causal relationships between a subset of observed variables—e.g., sex and age are not causally related—while missing the rest. When only partial knowledge on the graph is available—i.e., we only know the causal relationship between a subset of the observed variables, we can instead work with a modified acyclic graph $\tilde{\boldsymbol{A}}$ obtained by finding the strongly connected components as in [37], where subsets of variables with unknown causal relationships are treated as a block (see §7 for an example).

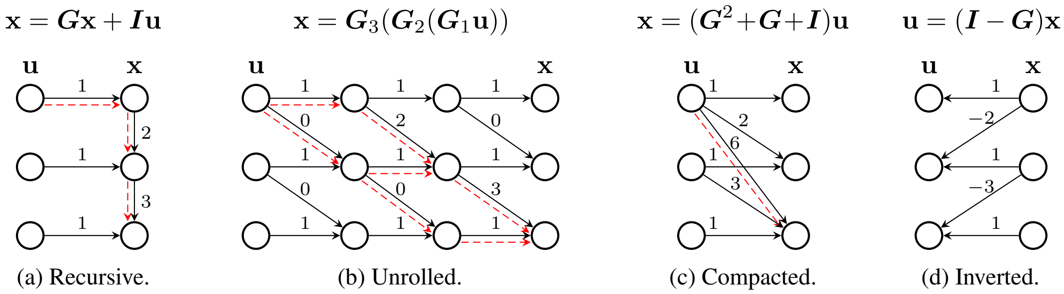


Figure 4: Example of the linear SCM $\{\mathrm{x}_1 := \mathrm{u}_1\,;\ \mathrm{x}_2 := 2\mathrm{x}_1 + \mathrm{u}_2\,;\ \mathrm{x}_3 := 3\mathrm{x}_2 + \mathrm{u}_3\}$ written (a) in its usual recursive formulation; (b) without recursions, with each step made explicit; (c) without recursions, as a single function; and (d) writing $\mathbf{u}$ as a function of $\mathbf{x}$. The red dashed arrows show the influence of u_1 on x_3 for all equations from $\mathbf{u}$ to $\mathbf{x}$, with the compacted version exhibiting shortcuts (see §4). Note that in the linear case we have $\boldsymbol{A} \coloneqq \boldsymbol{G} \neq \mathbf{0}$, and that $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3 \preceq \boldsymbol{G} + \boldsymbol{I}$ are any three matrices such that their product equals $\boldsymbol{G}^2 + \boldsymbol{G} + \boldsymbol{I}$.

This allows us to reuse our theoretical results for known parts of the graph, thus generalizing the *block identifiability* results from von Kügelgen et al. [38].

4 Effective design of causal normalizing flows

We showed in §3 that causal NFs are a natural choice to learn the underlying SCM generating the data. Importantly, Thm. 1 assumes that we can find a causal NF whose observational distribution perfectly matches the true data distribution (according to the underlying SCM). In practice, however, reaching the optimal parameters may be tricky as: i) we only have access to a finite amount of training data; and ii) the optimization process for causal NFs (like for any neural network) may converge to a local optima. In this section, we analyse different design choices for causal NFs to guide the optimization towards solutions that do not only provide an accurate fit of the observational distribution, but allow us to also accurately answer to interventional and counterfactual queries.

Let us start with an illustrative example. Suppose that we are given the linear SCM in Fig. 4a, and we want to write the SCM equations as a TMI map to approximate them with a causal NF. As discussed in §3, we can unroll the causal equations (Fig. 4b)—resulting in a composition of functions structurally as sparse as $\boldsymbol{I} + \boldsymbol{A}$. These functions can be compacted into a single transformation (Fig. 4c), such that each x_i depends on its ancestors, $\mathbf{u}_{\mathrm{an}_i}$. However, note that in this step *shortcuts* appear, making direct and indirect causal paths in this representation indistinguishable—in our example, the indirect causal path from u_1 to x_3 present in Fig. 4a and Fig. 4b does not go anymore through the path that generates x_2, but instead via a *shortcut* that directly connects u_1 to x_3. Alternatively, we can invert the equations to write $\mathbf{u}$ as a function of $\mathbf{x}$ (Fig. 4d), which is structurally equivalent to $\boldsymbol{I} - \boldsymbol{A}$.

We remark that the above steps can be applied to any considered acyclic SCM (refer to App. B for a more detailed discussion). In particular, we can unroll the equations in a finite number of steps, and we can similarly reason about the causal dependencies through the Jacobian matrices of the generators, $\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x})$ and $\nabla_{\mathbf{u}} T_{\boldsymbol{\theta}}^{-1}(\mathbf{u})$. Moreover, note that the diffeomorphic assumption implies that we can invert the causal equations. Next, inspired by the different representations of an SCM (exemplified in Fig. 4), we consider the following design choices for causal NFs:

Generative model The first architecture imitates the unrolled equations (Fig. 4b), i.e., the causal NF is defined as a function from $\mathbf{u}$ to $\mathbf{x}$. Importantly, when full knowledge of the causal graph $\boldsymbol{A}$ is assumed, we also replicate the structural sparsity per layer by adequately masking the flow with $\boldsymbol{I} + \boldsymbol{A}$. In this way, the information from $\mathbf{u}$ to $\mathbf{x}$ is restricted to flow as if we were unrolling the causal model [36] and, as a result, the output of the $l-1$-th layer of the causal NF is given by:[3]

$$\mathrm{z}_i^{l-1} = \tau_i(\mathrm{z}_i^l; \mathbf{h}_i^{l-1}), \quad \text{where} \quad \mathbf{h}_i^{l-1} = c_i(\mathbf{z}_{\mathrm{pa}_i}^l). \tag{5}$$

Note that, by restricting each layer such that $\nabla_{\mathbf{z}^l} \boldsymbol{\tau}(\mathbf{z}^l) \equiv \boldsymbol{I} + \boldsymbol{A}$, there cannot exist shortcuts at the optima. For example, in Fig. 4, the (indirect) information of u_1 to generate x_3 by first generating

[3]The order has been reversed w.r.t. the causal NF definition from Eq. 3.

x_2 needs to go through the middle nodes in Fig. 4b. However, as shown by Sánchez-Martin et al. [36], we need at least $L = \text{diam}(\boldsymbol{A})$ layers in our causal NF to avoid shortcuts and thus differentiate between the different direct and indirect causal paths connecting a pair of observed variables.

In contrast, if we only know the causal ordering, then the causal NF will need to rule out the spurious correlations by learning during training the necessary zeroes to fulfil causal consistency (Cor. 2), i.e., such that $\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x}) \equiv \boldsymbol{I} + \boldsymbol{A}$ and $\nabla_{\mathbf{u}} T_{\boldsymbol{\theta}}^{-1}(\mathbf{u}) \equiv \boldsymbol{I} + \sum_{n=1}^{\text{diam}(\boldsymbol{A})} \boldsymbol{A}^n$.

Abductive model Reminiscent to the abduction step [32], another natural choice is to model the inverse equations of the SCM as in Fig. 4d, hence building a causal NF from $\mathbf{x}$ to $\mathbf{u}$. Under a known causal graph, we can again use extra masking in the causal NF to force each layer l to be structurally equivalent to $\boldsymbol{I} - \boldsymbol{A}$, such that

$$z_i^l = \tau_i(z_i^{l-1}; \mathbf{h}_i^l) , \quad \text{where} \quad \mathbf{h}_i^l = c_i(\mathbf{z}_{\text{pa}_i}^{l-1}) . \tag{6}$$

Remarkably, this architecture is capable of capturing all indirect dependencies of $\mathbf{u}$ on $\mathbf{x}$, even with a single layer. This is a result of the autoregressive nature of the ANFs used here to build causal NFs, as they compute the inverse sequentially. In the example of Fig. 4, the indirect influence of u_1 on x_3 via x_2 has to necessarily generate x_2 first (Fig. 4a). Similar to the previous architecture, in the absence of a causal graph (i.e., when only the causal ordering is known), the causal NF will need to rely on optimization to discard all spurious correlations.

4.1 Necessary conditions

We next analyse the necessary conditions for the design of a causal NF to be able to accurately approximate and manipulate an SCM. A summary of the analysis can be found in Tab 1.

Expressiveness The least restrictive condition is that the causal NF should be able to reach the optima and, as mentioned in §3, a single ANF layer (Eq. 3) is a universal TMI approximator [27].

Identifiability In order to perform interventions as we describe later in §5, we need the causal NF to isolate the exogenous variables, so that we can associate them with their respective endogenous variables. As we saw in §3, if the causal NF is expressive enough, and *if it follows a valid causal ordering w.r.t. the true causal graph* $\boldsymbol{A}$, then Thm. 1 ensures that we can isolate the exogenous variables up to elementwise transformations.

Causal consistency As stated in Cor. 2, the causal NF needs to share the causal dependencies of the SCM at the optima, meaning that their Jacobian matrices need to be structurally equivalent, i.e., $\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x}) \equiv \boldsymbol{I} - \boldsymbol{A}$ (Fig. 4d), and $\nabla_{\mathbf{u}} T_{\boldsymbol{\theta}}^{-1}(\mathbf{u}) \equiv \sum_{n=1}^{\text{diam}(\boldsymbol{A})} \boldsymbol{A}^n + \boldsymbol{I}$ (Fig. 4c). Given the (partial) causal graph $\boldsymbol{A}$, the generative model in Eq. 5 by design holds $\nabla_{\mathbf{u}} T_{\boldsymbol{\theta}}^{-1}(\mathbf{u}) \equiv \sum_{n=1}^{\text{diam}(\boldsymbol{A})} \boldsymbol{A}^n + \boldsymbol{I}$ for any sufficient number of layers $L \geq \text{diam}\, \boldsymbol{A}$ (see [36, Prop. 1]), however, there might still exist spurious paths from $\mathbf{x}$ to $\mathbf{u}$. Similarly, the abductive model in Eq. 6, while may not remove all spurious paths from $\mathbf{x}$ to $\mathbf{u}$ if $L > 1$, ensures causal consistency when $L = 1$. In cases where the selected architecture for the causal NF does not ensure causal consistency by design, but we have access to the causal graph, we can use this extra information to regularize our MLE problem as

$$\underset{\boldsymbol{\theta}}{\text{minimize}} \quad \mathrm{E}_{\mathbf{x}} \left[-\log p(T_{\boldsymbol{\theta}}(\mathbf{x})) + \| \nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x}) \odot (\mathbf{1} - \boldsymbol{A}) \|_2 \right] , \tag{7}$$

where $\mathbf{1}$ is a matrix of ones, thus penalizing spurious correlations from $\mathbf{x}$ to $T_{\boldsymbol{\theta}}(\mathbf{x})$.

Tab 1 summarizes the discussed properties of the considered design choices. Remarkably, the abductive model with a single layer (similar to Fig. 4d) *enjoys all the necessary properties of a causal NF by design*. That is, the abductive model with $L = 1$ is expressive, and causally consistent w.r.t. the provided causal graph $\boldsymbol{A}$, greatly simplifying the optimization process.

Remark. It is not straightforward why abductive models can be causally consistent by design, but generative models cannot. The answer lies on the structure of the problem. Intuitively, this is a consequence of the mapping $\mathbf{x} \to \mathbf{u}$ being structurally sparser (u_i depends on pa_i, see Fig. 4d) than that of $\mathbf{u} \to \mathbf{x}$ (x_i depends on an_i, see Fig. 4c). Therefore, ensuring causal consistency from $\mathbf{u}$ to $\mathbf{x}$ does not necessarily imply causal consistency from $\mathbf{x}$ to $\mathbf{u}$.

Table 1: Summary of the considered design choices, their induced properties, and their time complexity for density evaluation and sampling. Generative models design their forward pass as $\mathbf{u} \to \mathbf{x}$, and abductive models as $\mathbf{x} \to \mathbf{u}$. See §4 for an in-depth discussion.

Design Choices			Model Properties		Time Complexity	
Network Type		Causal Asumption	Causal Consistency		Sampling	Evaluation
			$\mathbf{u} \to \mathbf{x}$	$\mathbf{x} \to \mathbf{u}$		
$\mathbf{u} \to \mathbf{x}$	Generative	Ordering	✗	✗	$\mathcal{O}(L)$	$\mathcal{O}(dL)$
$\mathbf{u} \to \mathbf{x}$	Generative	Graph $\boldsymbol{A}$	✓	✗	$\mathcal{O}(L)$	$\mathcal{O}(dL)$
$\mathbf{x} \to \mathbf{u}$	Abductive	Ordering	✗	✗	$\mathcal{O}(dL)$	$\mathcal{O}(L)$
$\mathbf{x} \to \mathbf{u}$	Abductive ($L > 1$)	Graph $\boldsymbol{A}$	✗	✗	$\mathcal{O}(dL)$	$\mathcal{O}(L)$
$\mathbf{x} \to \mathbf{u}$	Abductive ($L = 1$)	Graph $\boldsymbol{A}$	✓	✓	$\mathcal{O}(dL)$	$\mathcal{O}(L)$

5 Do-operator: enabling interventions and counterfactuals

In this section, we propose an implementation of the *do-operator* well-suited for causal NFs, such that we can evaluate the effect of interventions and counterfactuals [30]. The *do-operator* [32], denoted as $do(\mathrm{x}_i = \alpha)$, is a mathematical operator that simulates a physical intervention on an SCM $\mathcal{M}$, inducing an alternative model $\mathcal{M}^{\mathcal{I}}$ that fixes the observational value $\mathrm{x}_i = \alpha$, and thus removes any causal dependency on x_i. Usually, the do-operator is implemented by yielding an SCM $\mathcal{M}^{\mathcal{I}} = (\tilde{\mathbf{f}}^{\mathcal{I}}, P_{\mathbf{u}})$ result of replacing the i-th component of $\tilde{\mathbf{f}}$ with a constant function, $\tilde{f}_i^{\mathcal{I}} := \alpha$. Unfortunately, this implementation of the do-operator only works for the recursive representation of the SCM (Fig. 4a), thus not generalizing to the different architecture designs of causal NFs discussed in §4 the previous section.

We instead propose to manipulate the SCM by modifying the exogenous distribution $P_{\mathbf{u}}$, while keeping the causal equations $\tilde{\mathbf{f}}$ untouched. Specifically, an intervention $do(\mathrm{x}_i = \alpha)$ updates $P_{\mathbf{u}}$, restricting the set of plausible $\mathbf{u}$ to those that yield the intervened value α. We define the intervened SCM as $\mathcal{M}^{\mathcal{I}} = (\tilde{\mathbf{f}}, P_{\mathbf{u}}^{\mathcal{I}})$, where the density of $P_{\mathbf{u}}^{\mathcal{I}}$ is of the form

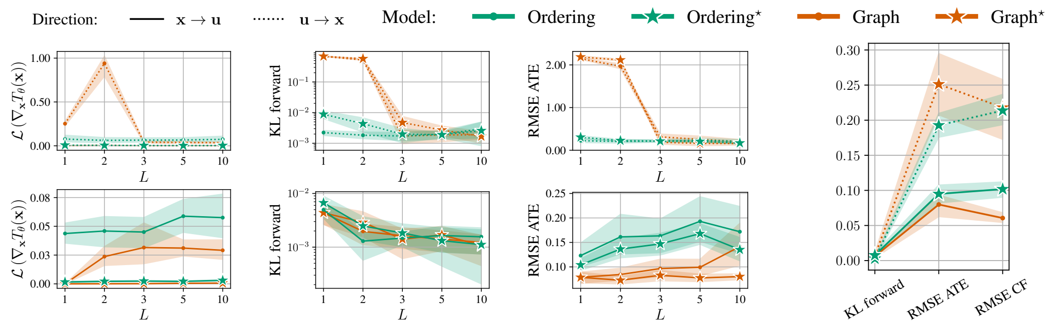
$$p^{\mathcal{I}}(\mathbf{u}) = \delta\left(\left\{\tilde{f}_i(\mathbf{x}_{\mathrm{pa}_i}, \mathrm{u}_i) = \alpha\right\}\right) \cdot \prod_{j \neq i} p_j(\mathrm{u}_j), \tag{8}$$

and where δ is the Dirac delta located at the unique value of u_i that yields $\mathrm{x}_i = \alpha$ after applying the causal mechanism $\tilde{f}_i$. This approach resembles the one proposed for soft interventions [11] and backtracking counterfactuals [39]. Moreover, as shown in App. C, it can be generalized for non-bijective causal equations. Note also that Eq. 8 is well-defined only if the set $\{\mathbf{u} \sim P_{\mathbf{u}} \,|\, \tilde{\mathbf{f}}_i(\boldsymbol{x}, \mathbf{u}) = \alpha\}$ is non-empty, i.e., if the intervened variable takes a *plausible* value (i.e., with positive density in the original causal model). Remarkably, this implementation works directly on the distribution of the exogenous variables and can be applied to any SCM representation (see Fig. 4), and therefore any of the architectures for the causal NF in §4.

Implementation details We take advantage of the autoregressive nature of causal NFs, and generate samples from Eq. 8 by: i) obtaining the exogenous variables $\mathbf{u} := T_{\boldsymbol{\theta}}(\mathbf{x})$; ii) replacing the observational value by its intervened value, $\mathrm{x}_i := \alpha$; and iii) by setting u_i to the value of the i-th component of $T_{\boldsymbol{\theta}}(\mathbf{x})$. If the causal NF has successfully isolated the exogenous variables (§3), and it preserves the true causal paths (§4), then the causal NF ensures that $\mathrm{x}_i = \alpha$ independently of the value of its ancestors, since u_i can be seen in Eq. 8 as a deterministic function of the given α and the value of its parents (and therefore its ancestors). We provide further details and the step-by-step algorithms to compute interventions and counterfactuals with causal NFs in App. C.

6 Empirical evaluation

In this section, we empirically validate the insights from §4, and compare causal NFs with previous works. Additional results and in-depth descriptions can be found in App. D.



(a) Ablation study on a 4-node chain SCM, as we vary the number of layers L. Top: generative models ($\mathbf{u} \to \mathbf{x}$). Bottom: abductive models ($\mathbf{x} \to \mathbf{u}$).

(b) Comparison of the two best models of each row.

Figure 5: Ablation of different choices of the causal NF to be causally consistent, and capture the observational and interventional distributions. The use of regularization on the Jacobian (Eq. 7) is indicated with the $\star$ superscript. The abductive causal NF with information on $\boldsymbol{A}$ and $L = 1$ outperforms the rest of models across all metrics, demonstrating its efficacy and simplicity.

6.1 Ablation study

Experimental setup We evaluate every network combination described in Tab 1 on a 4-chain SCM (which has diameter 3 and a very sparse Jacobian) and assess the extent to which these models: i) capture the observational distribution, using $\mathrm{KL}(p_{\mathcal{M}} \| p_{\boldsymbol{\theta}})$; ii) remain causally consistent w.r.t. the original SCM, measured via $\mathcal{L}(\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x})) \coloneqq \|\nabla_{\mathbf{x}} T_{\boldsymbol{\theta}}(\mathbf{x}) \cdot (\mathbf{1} - \boldsymbol{A})\|_2$ from Eq. 7; and iii) perform at interventional tasks, such as estimating the Average Treatment Effect (ATE) [31] and computing counterfactuals, both of which we measure with the RMSE w.r.t. the original SCM. Every experiment is repeated 5 times, and every causal NF uses Mask Autoregressive Flows (MAFs) [26] as layers.

Results Fig. 5 shows the result for different design choices of causal NFs. Specifically, we show: network design (generative $\mathbf{u} \to \mathbf{x}$ vs. abductive $\mathbf{x} \to \mathbf{u}$), causal knowledge (ordering vs. graph), number of layers L, and whether to use MLE with regularization (Eq. 4 vs. 7).

First, we see in Fig. 5a (top) that, as expected, the generative models ($\mathbf{u} \to \mathbf{x}$) using the causal graph cannot capture the SCM with $L < \operatorname{diam} \boldsymbol{A} = 3$. Furthermore, we observe that abductive models $\mathbf{x} \to \mathbf{u}$ (Fig. 5a, bottom) accurately fit the observational distribution, and that embedding the causal graph in the architecture significantly improves the ATE estimation.

Second, we now compare the two network designs in Fig. 5b, and observe that in general abductive models results in more accurate estimates of the observational distribution, as well as of interventional and counterfactual queries. Finally, we observe that regularization works well in all cases, yet it renders useless for the abductive model with $L = 1$ and knowledge on the graph, since it is causally consistent by design. In summary, our experiments confirm that, despite its simplicity, the causal abductive model with $L = 1$ outperforms the rest of design choices. As a consequence, in the following sections we will stick to this particular design choice, and refer to it as causal NF.

6.2 Non-linear SCMs

Experimental setup We compare our causal NF (causal, abductive, and with $L = 1$) with two relevant works: i) CAREFL [18], an abductive NF with knowledge on the causal ordering and affine layers; and ii) VACA [36], a variational auto-encoding GNN with knowledge on the graph. For fair comparison, every model uses the same budget for hyperparameter tuning, our causal NF uses affine layers, and CAREFL has been modified to use the proposed do-operator from §5 (as the original implementation only works in root nodes). We increase the complexity of the SCMs and consider: i) TRIANGLE, a 3-node SCM with a dense causal graph; ii) LARGEBD [13], a 9-node SCM with non-Gaussian $P_{\mathbf{u}}$ and made out of two chains with common initial and final nodes; and iii) SIMPSON [13], a 4-node SCM simulating a Simpson's paradox [35], where the relation between two variables changes if the SCM is not properly approximated.

Results The results are summarized in Tab 2. In a nutshell: *the proposed causal NF outperforms both CAREFL and VACA in terms of performance and computational efficiency.* VACA shows poor performance, and is considerably slower due to the complexity of GNNs. Our causal NF outperforms

Table 2: Comparison, on three non-linear SCMs, of the proposed causal NF, VACA [36], and CAREFL [18] with the do-operator proposed in §5. Results averaged over five runs.

		Performance			Time Evaluation (μs)		
Dataset	Model	KL	ATE_{RMSE}	CF_{RMSE}	Training	Evaluation	Sampling
TRIANGLE NLIN [36]	Causal NF	$\mathbf{0.00_{0.00}}$	$\mathbf{0.12_{0.03}}$	$\mathbf{0.13_{0.02}}$	$\mathbf{0.52_{0.07}}$	$\mathbf{0.58_{0.07}}$	$\mathbf{1.07_{0.12}}$
	CAREFL†	$\mathbf{0.00_{0.00}}$	$\mathbf{0.12_{0.03}}$	$0.17_{0.03}$	$\mathbf{0.57_{0.18}}$	$\mathbf{0.83_{0.26}}$	$\mathbf{1.68_{0.62}}$
	VACA	$7.71_{0.60}$	$4.78_{0.01}$	$4.19_{0.04}$	$28.82_{1.21}$	$23.00_{0.55}$	$70.65_{3.70}$
LARGEBD NLIN [13]	Causal NF	$\mathbf{1.51_{0.04}}$	$\mathbf{0.02_{0.00}}$	$\mathbf{0.01_{0.00}}$	$\mathbf{0.52_{0.10}}$	$\mathbf{0.60_{0.17}}$	$\mathbf{3.05_{0.66}}$
	CAREFL†	$1.51_{0.05}$	$0.05_{0.01}$	$0.08_{0.01}$	$0.84_{0.47}$	$1.18_{0.17}$	$8.25_{1.29}$
	VACA	$53.66_{2.07}$	$0.39_{0.00}$	$0.82_{0.02}$	$164.92_{11.10}$	$137.88_{15.72}$	$167.94_{25.75}$
SIMPSON SYMPROD [13]	Causal NF	$\mathbf{0.00_{0.00}}$	$\mathbf{0.07_{0.01}}$	$\mathbf{0.12_{0.02}}$	$\mathbf{0.59_{0.17}}$	$\mathbf{0.60_{0.11}}$	$\mathbf{1.51_{0.30}}$
	CAREFL†	$\mathbf{0.00_{0.00}}$	$0.10_{0.02}$	$0.17_{0.04}$	$\mathbf{0.49_{0.15}}$	$0.81_{0.19}$	$1.91_{0.33}$
	VACA	$13.85_{0.64}$	$0.89_{0.00}$	$1.50_{0.04}$	$49.26_{4.09}$	$37.78_{3.41}$	$79.20_{14.60}$

Table 3: Accuracy, F1-score, and counterfactual unfairness of the audited classifiers. Causal NFs enable both fair classifiers and accurate unfairness metrics. Results are averaged on five runs.

	Logistic classifier				SVM classifier			
	full	unaware	fair $\mathbf{x}$	fair $\mathbf{u}$	full	unaware	fair $\mathbf{x}$	fair $\mathbf{u}$
f1	$72.28_{6.16}$	$72.37_{4.90}$	$59.66_{8.57}$	$73.08_{4.38}$	$76.04_{2.86}$	$76.80_{5.82}$	$68.28_{5.74}$	$77.39_{1.52}$
accuracy	$67.00_{3.83}$	$66.75_{2.63}$	$54.75_{5.91}$	$66.50_{3.70}$	$69.50_{3.11}$	$71.00_{3.83}$	$59.25_{2.99}$	$69.75_{1.26}$
unfairness	$5.84_{2.93}$	$2.81_{0.72}$	$0.00_{0.00}$	$0.00_{0.00}$	$6.65_{2.45}$	$2.78_{0.40}$	$0.00_{0.00}$	$0.00_{0.00}$

CAREFL in counterfactual estimation tasks with identical observational fitting, showing once more the importance of being causally consistent. Even more, our causal NF is also quicker than CAREFL, as best-performing CAREFL architectures have in general more than one layer.

7 Use-case: fairness auditing and classification

To show the potential practical impact of our work, we follow the fairness use-case of Sánchez-Martin et al. [36] on the German Credit dataset [9]—a dataset from the UCI repository where the likelihood of individuals repaying a loan is predicted based on a small set of features, including sensitive attributes such as their sex. Extra details and results appear in App. E.

Experimental setup As proposed by Chiappa [4], we use a partial graph which groups the 7 discrete features of the dataset in 4 different blocks with known causal relationships, putting in practice the results from §3.1. For the causal NF, we use the abductive model with a single non-affine neural spline layer [10]. Our ultimate goal is to train a causal NF that captures well the underlying SCM, and use it to train and evaluate classifiers that predict the (additional) binary feature *credit risk*, while remaining counterfactually fair w.r.t. the binary variable *sex*, x_S.

In this setting, we call a binary classifier $\kappa : \mathbb{X} \to \{0, 1\}$ counterfactually fair [23] if, for all possible factual values $\mathbf{x}^{\mathrm{f}} \in \mathbb{X}$, the counterfactual unfairness remains zero. That is, if we have that $\mathrm{E}_{\mathbf{x}^{\mathrm{f}}}\left[P(\kappa(\mathbf{x}^{\mathrm{cf}}) = 1 \,|\, do(\mathrm{x}_S = 1), \mathbf{x}^{\mathrm{f}}) - P(\kappa(\mathbf{x}^{\mathrm{cf}}) = 1 \,|\, do(\mathrm{x}_S = 0), \mathbf{x}^{\mathrm{f}})\right] = 0$, where $\mathbf{x}^{\mathrm{cf}}$ is a counterfactual sample coming from the distribution $P(\mathbf{x}^{\mathrm{cf}} \,|\, do(\mathrm{x}_S = s), \mathbf{x}^{\mathrm{f}})$, for $s = 0, 1$.

Following Sánchez-Martin et al. [36], we audit: a model that takes all observed variables (*full*); an *unaware* model that leaves the sensitive attribute x_S out; a fair model that only considers non-descendant variables of x_S (*fair* $\mathbf{x}$); and, to demonstrate the ability to learn a counterfactually fair classifier, we include a classifier that takes $\mathbf{u} = T_{\boldsymbol{\theta}}(\mathbf{x})$ as input, but leaves u_S out (*fair* $\mathbf{u}$).

Results Tab 3 summarizes the performance and unfairness of the classifiers, using logistic regression [7] and SVMs [6]. Here, we observe that by taking the non-sensitive exogenous variables from the causal NF, the obtained classifiers achieve comparable or better accuracy than the rest of the classifiers, while at the same time being counterfactually fair. Moreover, the estimations of unfairness obtained with the causal NF match our expectations [23], with *full* being the most unfair, followed by *aware* and the two fair models. With this use-case, we demonstrate that *Causal NFs may indeed be a valuable asset for real-world causal inference problems.*

8 Concluding remarks

In this work, we have shown—both theoretically and empirically—that causal NFs are a natural choice to learn a broad class of causal data-generating processes in a principled way. Specifically, we have proven that causal NFs can match the observational distribution of an underlying SCM, and that in doing so the ANF needs to be causally consistent. However, as limited data availability and local optima may hamper reaching these solutions in practice, we have explored different network designs, exploiting the available knowledge on the causal graph. Moreover, we have provided causal NFs with a do-operator to efficiently solve causal inference tasks. Finally, we have empirically validated our findings, and demonstrated that our causal NF framework: i) outperforms competing methods; and ii) can deal with mixed data and partial knowledge on the causal graph.

Practical limitations Despite considering a broad class of SCMs, we have made several assumptions that, while being standard, may not hold in some application scenarios. With regard to our causal assumptions, the presence of unmeasured hidden confounders may break our causal sufficiency assumption; mismatches between the true causal graph (e.g., it may contain cycles) and our assumed graph/ordering may lead to poor estimates of interventional and counterfactual queries; and the non-bijective true causal dependencies may invalidate our theoretical and thus practical findings. Besides, we have focused on MLE estimation for learning the causal NF. However, MLE does not test the independency of the exogenous variables during training, which would also break our causal sufficiency assumption.

Future work We firmly believe that our work opens a number of interesting directions to explore. Naturally, we would like to address current limitations by, e.g., using interventional data to address the existence of hidden confounders [16, 25], explore alternative losses other than MLE (e.g., flow matching [24]). Moreover, it would be exciting to see causal NFs applied to other problems such as causal discovery [13], fair decision-making [23], or neuroimaging [14], among others. However, we would like to stress that, in the above contexts, it would be essential to validate the suitability of our framework (e.g., using experimental data) to prevent potential harms.

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